math unit 2 numerical integration

1] TRAPAZOIDAL RULE EXAMPLES

Questions:

1. $\int_0^2 (3x^2 + 2)\, dx$
2. $\int_1^4 (2x + 1)\, dx$
3. $\int_{-1}^1 e^x\, dx$
4. $\int_0^\pi \sin(x)\, dx$
5. $\int_1^3 \frac{1}{x}\, dx$
6. $\int_{-2}^2 x^3\, dx$
7. $\int_0^{\frac{\pi}{2}} \cos(x)\, dx$
8. $\int_{-1}^2 (x^2 + x + 1)\, dx$
9. $\int_1^4 \sqrt{x}\, dx$
10. $\int_0^1 e^{-x^2}\, dx$

value of n for Q. BY JAY...

1]- 4

2]- 4

3]-3

4]-2

5]-5

6]-3

7]-3

8]-2

9]-6

10]-4

Example 1:

$\int_0^2 (3x^2 + 2)\, dx$

Solution:

$a = 0, \quad b = 2, \quad n = 4$

$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5$

Compute $f(x_i)$ for $x_i = 0, 0.5, 1.0, 1.5, 2.0$:

$f(0) = 2, \quad f(0.5) = 2.75, \quad f(1.0) = 5, \quad f(1.5) = 10.25, \quad f(2.0) = 14$

Apply the Trapezoidal Rule formula:

$\text{Area} = \frac{\Delta x}{2}[f(a) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(b)]$

$\text{Area} = \frac{0.5}{2}[2 + 2(2.75) + 2(5) + 2(10.25) + 14]$

$\text{Area} = \frac{0.5}{2}[52] = 13$

Example 2:

$\int_1^4 (2x + 1)\, dx$

Solution:

$a = 1, \quad b = 4, \quad n = 4$

$\Delta x = \frac{b-a}{n} = \frac{4-1}{4} = 0.75$

Compute $f(x_i)$ for $x_i = 1, 1.75, 2.5, 3.25, 4$:

$f(1) = 3, \quad f(1.75) = 5.5, \quad f(2.5) = 8, \quad f(3.25) = 10.5, \quad f(4) = 13$

Apply the Trapezoidal Rule formula:

$\text{Area} = \frac{\Delta x}{2}[f(a) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(b)]$

$\text{Area} = \frac{0.75}{2}[3 + 2(5.5) + 2(8) + 2(10.5) + 13]$

$\text{Area} = \frac{0.75}{2}[54] = 20.25$

Final Answer:

$\int_1^4 (2x + 1)\, dx \approx 20.25$

SIMPSONS 1/3 RULE

1. **Question:**
   Evaluate $\int_0^1 x^2\,dx$ using Simpson's 1/3 Rule with 4 subintervals.
   Answer:
   Approximate Integral = 0.25
2. **Question:**
   Calculate $\int_{-1}^1 (3x^2 + 2x - 1)\,dx$ using Simpson's 1/3 Rule with 6 subintervals.
   Answer:
   Approximate Integral = 2.66667
3. **Question:**
   Find $\int_1^2 \sqrt{x}\,dx$ using Simpson's 1/3 Rule with 8 subintervals.
   Answer:
   Approximate Integral = 0.83333
4. **Question:**
   Evaluate $\int_0^\pi \sin(x)\,dx$ using Simpson's 1/3 Rule with 10 subintervals.
   Answer:
   Approximate Integral = 2.00000
5. **Question:**
   Calculate $\int_{-2}^2 e^x\,dx$ using Simpson's 1/3 Rule with 12 subintervals.
   Answer:
   Approximate Integral = 10.9325

6. **Question:**
   Find $\int_1^3 (x^3 + 2x^2 + 1)\, dx$ using Simpson's 1/3 Rule with 14 subintervals.
   Answer:
   Approximate Integral = 32.4286
7. **Question:**
   Evaluate $\int_0^{\frac{\pi}{2}} \cos(x)\, dx$ using Simpson's 1/3 Rule with 16 subintervals.
   Answer:
   Approximate Integral = 1.00000
8. **Question:**
   Calculate $\int_{-1}^1 \frac{1}{1+x^2}\, dx$ using Simpson's 1/3 Rule with 18 subintervals.
   Answer:
   Approximate Integral = 3.14159
9. **Question:**
   Find $\int_2^4 \ln(x)\, dx$ using Simpson's 1/3 Rule with 20 subintervals.
   Answer:
   Approximate Integral = 1.73448
10. **Question:**
    Evaluate $\int_{-3}^3 (x^4 + 3x^2 + 1)\, dx$ using Simpson's 1/3 Rule with 22 subintervals.
    Answer:
    Approximate Integral = 104.000

SIMPSONS' 3/8 RULE

1. **Question:**
   Evaluate $\int_1^5 (2x^3 - x^2 + 3)\, dx$ using Simpson's 3/8 Rule with 3 strips.
   Answer:
   Number of Strips: $n = 3$
   Final Answer: $\text{Approximate Integral} \approx 128.5$
2. **Question:**
   Calculate $\int_0^4 (x^2 + 2x + 1)\, dx$ using Simpson's 3/8 Rule with 4 strips.
   Answer:
   Number of Strips: $n = 4$
   Final Answer: $\text{Approximate Integral} \approx 30.6667$
3. **Question:**
   Find $\int_{-1}^2 e^x\, dx$ using Simpson's 3/8 Rule with 6 strips.
   Answer:
   Number of Strips: $n = 6$
   Final Answer: $\text{Approximate Integral} \approx 7.3453$
4. **Question:**
   Evaluate $\int_1^4 (4x^2 - 3x + 2)\, dx$ using Simpson's 3/8 Rule with 5 strips.
   Answer:
   Number of Strips: $n = 5$
   Final Answer: $\text{Approximate Integral} \approx 53.4$